AA-S-00275(GSA-FSS)
July 23, 1964

INTERIM FEDERAL SPECIFICATION

SHELVING, STORAGE AND DISPLAY, STEEL, BOLTLESS

This Interim Federal Specification was developed by the Standardization Division, Federal Supply Service, General Services Administration, Washington, D. C. 20407, based upon currently available technical information. It is recommended that Federal agencies use it for procurement and forward recommendations for changes to the preparing activity shown above.

1 Scope and Classification

1.1 Scope This specification describes knock-down type, steel storage and display shelving for merchandising of food at Government installations (Commissaries).

1.2 Classification

1.2 1 Types and sizes. The shelving units shall be of the following types and sizes as specified. (See 6.1.1).

Type I - Island

Shelves - 36 inch and 48 inch lengths.

	Overall widths (includes uprights)
Base shelf (width) - 16, 19, 22, 25, 28, 31 inches	34, 41, 46, 52, 58 and 64 inches
Second shelf (width) - 16, 19, 22, 25 inches	34, 40, 46 and 52 inches
Third shelf (width) - 13, 16, 19, 22 inches	28, 34, 40 and 46 inches
Fourth shelf (width) - 13, 16, 19, 22 inches	28, 34, 40 and 46 inches

Upright (posts) - 54, 60, 66 and 72 inches

Type II - Wall

Shelves - 36 inch and 48 inch lengths

	Overall widths (includes uprights)
Base shelf (width) - 16, 19, 22, 25, 28, 31 and 34 inches	18, 21, 24, 27, 30, 33 & 36 inches
Second shelf (width) - 16, 19, 22, 25 inches	18, 21, 24, 27, inches
Third shelf (width) - 13, 16. 19, 22 inches	15, 18, 21, 24 inches
Fourth shelf (width) - 13, 16, 19, 22 inches	15, 18, 21, 24 inches

Upright (posts) - 54, 60, 66, 72 and 84 inches.

2 APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards of the issues in effect on date of invitation for bids, form a part of this specification

Federal Specifications

LLL-B-00810 - Building Board. (Hardboard) Hard Pressed, Vegetable Fiber.
TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

Federal Standards:

Fed. Std. No. 102 - Preserving, Packaging and Packing Levels
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Use).

FSC 7125

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington D C , Atlanta, Chicago, Kansas City, Mo , Dallas Denver, San Francisco, Los Angeles, and Seattle, Wash

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies)

Military Standards.

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking for U S Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

2 2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Steel Products Manual.

(Copies may be obtained from the American Iron and Steel Institute, 350 Fifth Avenue, New York City, N. Y.)

Uniform Freight Classification Rules

(Application for copies may be addressed to the Uniform Freight Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois)

3. REQUIREMENTS

3.1 Preproduction sample.- Unless otherwise specified (see 6.1), prior to quantity production, preproduction samples of the finished commodity shall be made available at the contractor's plant or other designated location, for inspection and testing by a Government representative to determine compliance with the requirements of this specification The approved preproduction samples shall fix the design and construction features used on production units throughout the contract No variation will be permitted without written approval of the contracting officer

3 2 Material

3 2.1 Steel All steel sheet parts shall be cut, stamped or die-formed from FS-1010 - 1020 cold rolled steel sheets The steel shall be bright, smooth, free of scale, pits, wrinkles, scratches, laps, crimps and buckles affecting strength or appearance

3 2.2 Hardware.

3 2.2.1 Price Tag Molding - The price tag molding shall be attached full length to the front edge of all shelves and decks The molding shall be positive locking, of enameled steel or of extruded satin-finish aluminum and shall accept and retain, without sliding, standard 1-1/4 inch price tags Price tag molding shall extend 1/8 inch to 3/16 inch above front edge of shelves

3.2.2.2 Bolts, pins and other unexposed fastening devices shall be made rust resistant by electro-galvanizing or by zinc, chromium, or cadmium plating

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3.2.3 Hardboard.- Hardboard shall conform to Type IV, Surface 1, Finish A and Design a of LLL-B-00810.

3.2 4 Enamel - The enamel shall be of the baking type in accordance with the manufacturer's commercial product and shall meet the requirements of paragraph 3.4 2 and test specified in 4 4 2 The enamel or multicolor finish of upper face of shelving shall conform to the following when tested as specified in 4 4.2

Gloss 60° specular - 75 plus 10, minus .00
Simulated wear - strokes 150,000

3.2 4.1 Color of enamel - Color shall be as specified in the Invitation-for-Bid

3 3 Construction

3 3 1 Design.- The Type I and Type II shelving shall consist of free standing units not requiring fastening to walls or floor Shelving shall be of a design that facilitates rapid assembly or disassembly The unit section of shelving shall consist of a base shelf, intermediate shelves and uprights. (See Section 6)

3.3.2 Definition of terms used herein

3 3.2.1 Welding terms All spot welding and electrical resistance welding shall be as defined in Military Standard MIL-STD-20 Definitions of these terms in MIL-STD-20 are as follows·

Resistance welding A group of welding processes wherein coalescence is produced by the heat obtained from resistance of the work to the flow of electric current in a circuit of which the work is a part, and by the application of pressure

Spot Welding A resistance welding process wherein coalescence is produced by the heat obtained from resistance to the flow of electric current through the work parts held together under pressure by electrodes The size and shape of the individually formed welds are limited primarily by the size and contour of the electrodes.

Fusion welding and brazing shall be defined as follows

Fusion welding. Fusion welding consists of any welding process in which the weld is carried out solely by the melting of the metals to be joined, without any mechanical pressure The joining is usually accomplished through the fusion of metal deposits from an external source

Brazing· Brazing is the joining of metals with the aid of a filler metal of a different composition having a melting point above 800 degrees Fahrenheit Brazing method shall be gasflux, using nickel bronze rod (nickel content 9 5% tensile strength 70,000 - 75,000 p s.i.) acetylene, and oxygen (or propane) gases

3 3 2 2 Fabricating terms All bends, forms, seams and joints of the metal specified herein shall be as defined in Federal Standard No 187

3 3 3 Tolerances The following tolerances shall be permitted unless otherwise specified·

Width of shelves ± 1 inch
Length of upright ± 2 inches
Length of shelves ± 1/32 inch

3 3 4 Spot welding Spot welds shall not be closer to the edge of either adjoining member than 1/2 the greatest dimension of the spot weld, nor shall they be entered closer together than twice the greatest dimension. Squeeze time, weld time and hold time shall be controlled to produce uniform welds that will meet the test specified in 4 4 3 All exposed spot or projection welds shall be filled and ground smooth.

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3 3.5 Gages. - The minimum gage numbers for shelf parts are specified in Table I, and shall be as specified in the Manufacturers' Standard Gage for Steel Sheets, in the Steel Products Manual. The decimal equivalents shown for the steel sheet thicknesses in Table I are subject to the standard tolerances established in the Steel Products Manual.

TABLE I - Gages for shelving components

Part	Thickness (inch)	Gage
Shelves	0.0299	22
Shelf brackets	0.0747	14
Base brackets	0.1046	12
Toe panel (kick plate)	0.0299	22
Base trim	0.0359	20
Back panels	0.0359	20
End panels	0.0299	22
Uprights	0.1046	12
Shelf channels or hat sections	0.0299	22
End trim (base brackets)	0.0359	20

3 3 5 1 Base shelves. Steel top surfacing of shelf deck shall be of the gage specified in Table I The underside of shelf shall be reinforced by "U" shaped channels or hat sections approximately 2 inches in width and 1 inch in depth spaced about 6 inches apart The shelves shall be spot welded to meet the requirements specified in 3 3 4 with a sufficient number of spots to sustain loads listed in 4.4.1. Alternately, channels may be secured to shelving by other means except that shelves must sustain loads listed in 4 4.1.

3 3 5.2 Base Bracket - Base bracket shall be constructed of the gage of steel specified in Table I The bracket shall be designed with flanges or other formations to adequately support loads imposed upon the base deck The back end which engages the upright, shall be permanently attached to the upright in such a manner that it cannot be accidentally disengaged or loosen under load

3 3 5 2 1 End Trim - When furnished separately, end trim for base brackets may be of snap-on design or installed by means of screws or bolts

3 3 5 3 Base Front The base front shall interlock at both ends with base brackets or it may be of the free floating type that permits adjustment for irregularities in the floor surface The free floating type may be suspended from the ends and rise and fall vertically, or swing freely inward to permit cleaning under the base deck and provide a minimum of 3 inches setback

3 3 5 4 Shelf Brackets - Shelf brackets shall be stamped and formed of the gage of steel specified in Table I and permanently welded to the underside of the shelf ends or may be attached to the underside of shelf at time of assembly

3 3 5 4 1 Sloping shelves - When specified in the Invitation-for-Bids, sloping type shelf brackets or bracket and shelf assemblies having a downward angle of slope within the range of 15° to 20° shall be furnished

3.3.6 Adjustable intermediate shelves - Shelves shall be fabricated of the gage of steel specified in Table I When specified (see 6 2) steel shelves shall be surfaced with hardboard specified in 3.2 3. Hardboard employed as shelf surfacing may be attached to top of enameled steel shelf panel by fasteners or by adhesive

3 3.6 1 Wire dump shelves - When specified, intermediate shelves, 3 3 6, may be modified by addition of wire basket fronts and dividers In lieu of attachments to existing intermediate shelves, wire baskets or wire shelves may be substituted

3 3.7 Uprights - Uprights shall be constructed of hot or cold rolled steel ranging from 12 gage to 10 gage designed to support fully loaded shelves and meet the test requirements listed in paragraph 4.4.1 Uprights shall be perforated or slotted on 1 inch to 2 inch centers to permit required shelf height adjustment The underside of the upright shall be fitted with a leveling glide having a minimum height adjustment of 1 inch.

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3 3 8 Back panels.

3.3.8.1 Hardboard - Back panels shall consist of single or double (inner wood frame) panels of 1/4 inch tempered hardboard specified in 3 2 3 The tempered hardboard may be faced with a decorative laminate or furnished in natural color and finish. Metal cap mold shall be provided across top edge

3 3 8.2 Back panels, steel - Panels shall consist of a single sheet of steel supported by uprights and metal cap mold encasing top edge of sheet Gage of metal shall be as specified in Table I

3 3 9 End platforms, wall and island - End platform base sections shall be of construction similar to that of base shelves Platform section shall provide for adjustable and removable shelves of plywood, particle board or metal shelf construction similar to that employed for intermediate shelves Shelves shall be supported by one or more uprights Base sections shall be furnished in lengths necessary to span overall widths listed in 1.2 1 Shelf widths shall be as follows·

Base shelf - 15, 18, 21, and 24 inches
Intermediate shelf - 12, 15, 18 and 21 inches.

3 3 10 Assembly and erection - Use of bolts in the assembly of the components of the shelving sections shall be held to a minimum Bolts are permissible to tie individual free standing sections one to another. Individual shelves shall be of such design that they may be adjusted vertically without the removal of other shelves, or components. The vertical adjustment of shelves shall be on 1 to 2 inch centers It is recommended that 84 inch uprights be secured to wall when used with wall units having narrow width bases.

3.4 Pretreatment and finishing

3 4 1 Pretreatment - All exterior and interior unplated ferrous metal surfaces, shall be treated for painting in accordance with any method of Specification TT-C-490

3.4.2 Enameling. - All primed surfaces of the shelving shall be coated with enamel or multicolor finish as specified in 3.2.4 and shall be baked at a time-temperature cycle recommended by the paint manufacturer Total dry film thickness of enamel and primer shall be not less than 1 4 mils on all "wear areas' (exposed surfaces of steel shelves and uprights) and not less than 1.0 mil on other surfaces, inside and outside of the shelving Primer may be omitted from any surface prepared for painting in accordance with type I or type II of Specification TT-C-490 provided the dry enamel or multicolor finish film thickness is not less than 1.4 mils on all "wear areas', and not less than 0.6 mils on all other surfaces The finish shall level out to produce smooth, uniform surfaces without runs, wrinkles, grits, areas of thin film or no film, and separation of color

3 5 Erection instructions - Unless otherwise specified, printed erection instructions giving full and clear information for the erection of shelving specified in the contract shall be furnished by the supplier in the quantity specified.

3 6 Identification marking - Unless otherwise specified upper end of back panels shall be permanently marked in a plain and legible manner with the specification number, the Federal Stock Number, the date of manufacture, and the contractor's name or trade-mark, and the contract order number, so that the source of supply may be readily determined When specified, identification marking shall be in accordance with Military Standard MIL-STD-130

3.7 Workmanship - The shelves shall be constructed in strict accordance with the detailed requirements specified herein and be of commercial design The edges of all parts and sheets shall be further protected by folding, beading, flanging, or grinding to eliminate possible injury from burrs or rav edges The shelving components shall be free from any other defects which might affect durability, serviceability, or appearance

4 QUALITY ASSURANCE PROVISION

4 1 Inspection responsibility - The supplier is responsible for the performance of all inspection requirements as specified herein Except as other wise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

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4 2 Sampling for inspection and acceptance - Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in Standard MIL-STD-105, except where otherwise indicated

4.2.1 Component and material inspection.-In accordance with 4 1 above, the supplier is responsible for insuring that components and materials used are manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or if none, in accordance with this specification.

4.2 1.1 Process examination - Examination shall be made of the following operations to establish that no deviation is made from the indicated requirements Wherever a deviation is noted, correction shall be made Failure to make immediate correction may cause rejection of affected end item lots

a. Pretreatment and priming before finish material is applied as specified in 3.4.1, and 3.4.2.

b. Baking time of enamel as specified in 3.4.2

c Method of application, drying time, and temperature conditions of finishes as specified in 3.4.3.

4.2.2 End item inspection. -

4.2.2.1 Visual examination. - Examination of the shelving shall be in accordance with the classification of defects and acceptable quality levels (AQL's) specified in Table II with acceptable quality levels as specified in 4.2.2.3 The lot size shall be expressed in units of completely erected sections of steel shelving of the respective kind, type, class, style and size specified in the contract or order and shall be examined for the following defects·

TABLE II - Classification of defects in accordance with MIL-STD-105

Examine	Defect	Classification Major	Minor
Finish	Rust or indication of rust under paint	X	
	Not type or color specified		X
	Poor adhesion, i e peeling, blistered or flaky	X	
	Tacky or soft runs or sags		X
	Area of no film or thin film		X
	Oil under enamel	X	
	Not smooth or uniform		X
	Touchup not neat		X
	Discoloration, stain, or dirt		X
	Color separation		X
	Foreign matter imbedded in paint		X
	Color shade varies between components		X
	Average thickness of film less than specified	X	
Construction and workmanship	Not specified kind, type, class, style or size	X	
	Any component part not as specified		X
	Any component broken, fractured or cracked	X	
	Any component malformed, buckled, warped, bent out of shape or damage, affecting serviceability	X	

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TABLE II - Classification of defects in accordance with MIL-STD-105 (Continued)

Examine	Defect	Classification Major	Classification Minor
Construction and workmanship (cont'd)	Any component malformed, buckled, warped, bent out of shape or damaged, but not affecting serviceability		X
	Bolthole missing or in wrong location	X	
	panel visibly out of square	X	
	Any flange omitted		X
	Any flange not formed at right angle as specified		X

4.2.2.2 Inspection levels and AQL's.- The acceptable quality levels (AQL's) expressed in defects per hundred units and inspection levels shall be as follows:

	Inspection levels	AQL's
For examination in Table II Major	II	2.5
For examination in Table II Total	II	10.0
For examination in 4.2.2.2 (1 class)		4.0

4.2.2.3 Dimensional examination.- The units assembled for visual examination (see 4.2.2.2) shall be examined for defects in dimensions Any deviation from specification requirements shall constitute a defect The inspection level shall be S-2 with an AQL of 4.0.

4.2.2.4 Examination of preparation for delivery.- An examination will be made to determine that packaging, pack and marking, as required by section 5 of this specification, are complied with. The sample unit shall be one shipping container. The lot size shall be the number of containers in the inspection lot.

4.4 Tests - The lot size for testing shall be expressed in units of individual shelves. One sample unit shall be taken from every 250 shelves offered for inspection. Any failure in the test shall be cause for rejection of the lot.

4.4.1 Load test.- The load test shall be performed as specified herein.

Material Common Bricks.- 2-5/16 x 3-11/16 x 8-3/16 inches.

Equipment Tape or other linear measuring device.

Test specimen: Assembled 3 ft. or 4 ft. shelf unit, 1 each - wall and island, consisting of base shelf, uprights and three (3) adjustable shelves of the size on contract or order for the wall units, and six (6) adjustable shelves for the island unit.

Procedure Assemble three - 3 ft. or 4 ft shelf units as a single run of shelving Load the shelves in each section uniformly with a sufficient number of bricks to equal the specified load requirement for the shelf under test as given in Table III After the shelves are loaded, make the following measurements on the center unit:

a. Deflection at the center of the base shelf and the adjustable shelf.

b. Change in distance of front edge of the shelf bracket from floor from unloaded position.

c. Horizontal deflection of two center uprights.

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TABLE III

Shelf	Load, pounds	Deflection, Inches (Max)		
		a	b	c
se				
" and smaller	500	1/8	-	3/8
" - 31"	700	1/8	-	1
justable				
" and smaller	500	1/16	5/16	7/16
" - 25"	700	1/8	5/16	1
" - 31"	700	1/8	5/16	1

4 4 2 Test for enamel finish

4 4.2 1 Gloss 60 degree specular gloss shall be determined in accordance with Method 6101 of deral Standard No 141 using samples of finished shelving.

4 4 2.2 Simulated wear - Samples of finished shelving shall be subject to 150,000 twelve inch rokes with a No 10 can weighted to 10 pounds The surface of the shelving shall be examined r evidence of removal of enamel The appearance of bare metal in the stroked area shall constitute ilure of the enamel under test.

4 4 3 Spot welds. Spot welds shall meet the peel test specified in 4 1 4 of Specification L-W-12332 (ORD) and button diameter shall conform to Table II as specified therein

4 4 4 Bend test for steel sheets - The steel sheets used for parts to be formed shall withstand 180° bend at room temperature with bend across the grain without showing any crystallization, acks or fractures on the outside of the bend.

5 PREPARATION FOR DELIVERY

(Federal Standard No 102 shall be referred to for definitions and application of the various levels of packaging and packing for civilian procurements).

5 1 Preservation and packaging - Preservation and packaging shall be level A or C as specified see 6 1)

5 1.1 Level A - Packaging shall be as specified in the invitation for bids

5.1.2 Level C - Shelving shall be packaged according to manufacturers regular commercial ractice to assure safe delivery of goods at destination

5.2 Packing - Packing shall be level A, B, or C, as specified.

5.2.1 Level A - Packing shall be as specified in the invitation for bids

5.2.2 Level B - Packing shall be as specified in the invitation for bids

5.2 3 Level C - Shelving preserved and packaged as specified in 5 1 shall be packed in a nner to insure carrier acceptance and safe delivery at destination Containers shall be in accordance th Uniform Freight Classification Rules or regulations or other carriers applicable to the mode of ransportation.

5.3 Marking.

5 3.1 Civilian agencies - In addition to any special marking required by the contract or order, hipping containers shall be marked in accordance with Federal Standard No 123 and as specified in 3.3.

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5 3.2 Military requirements.- In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Standard MIL-STD-129 and as specified in 5.3 3

5.3 3 Additional marking.- In addition to the marking specified herein, arrows or prominent marks shall be placed on the sides of each shipping container to indicate the position in which the container shall be handled during shipment The word "UP" shall be placed at the point of the arrows in letters at least 2 inches in height.

6. NOTES:

6.1 Definitions (see section 5).- Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment

6.1.1 Ordering data.- Procurement documents should specify the following·

a. Title, symbol, and date of this specification
b Type required, and style or size, if applicable (see 1.2.1).
c. Width or depth of base shelving required (see 1.2.1)
d. Width and depth of intermediate shelving required (see 1.2.1).
e Height of uprights required
f Whether enameled steel surfaced shelving or hardboard surfaced shelving is required
g Whether downward sloping shelves are required (see 3 3.5.4).
h Color of enamel desired.
i Whether metal or double walled hardboard panel is required
j. Whether packaging and packing levels A, B, or C is required (see 5.1 and 5.2).

6.2 Intended use - The "boltless' type steel shelving covered by this specification is for use by commissaries of the military services. The specification includes basic shelving of the wall and island type as well as end platforms for display of merchandise. Shelving 36 and 48 inches in width, of various depths, is included in the specification Two foot lengths of shelving designed for fitting to columns and other obstructions are not covered by this specification

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